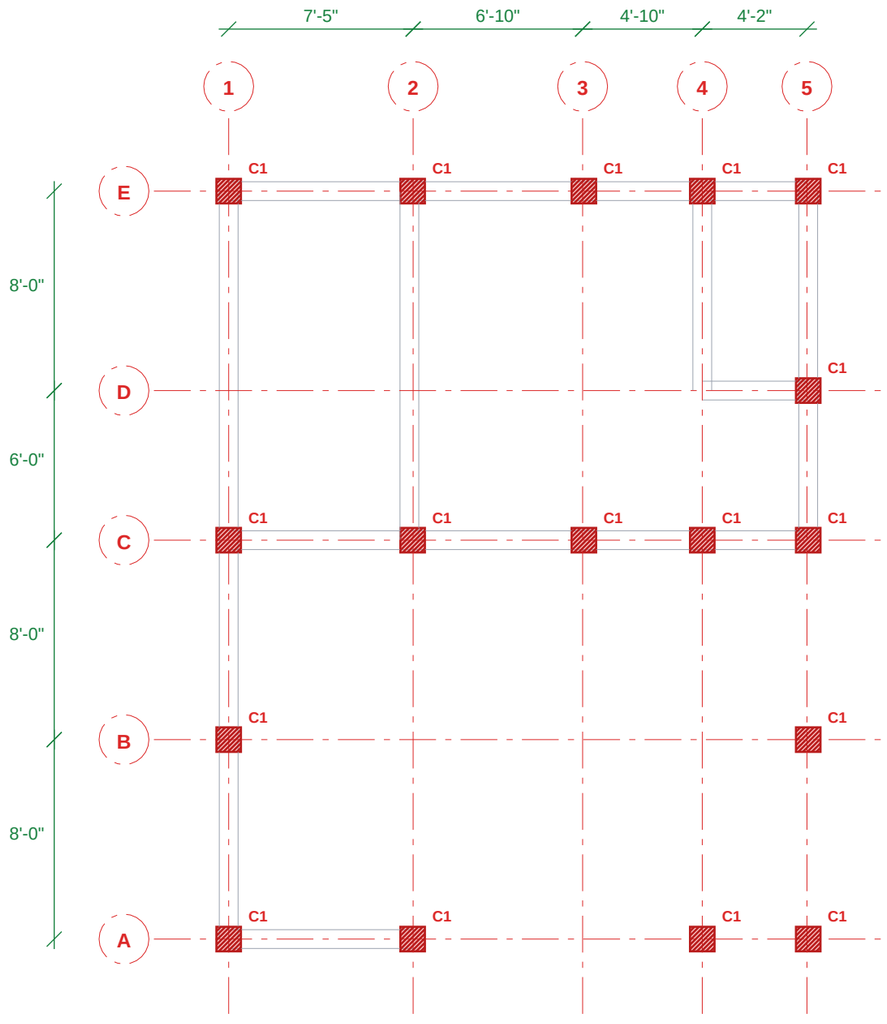
Column Layout & Grid



COLUMN LAYOUT & GRID (all floors)

Praganya Construction

Proprietor: Nishanth S · 98xxxxxx

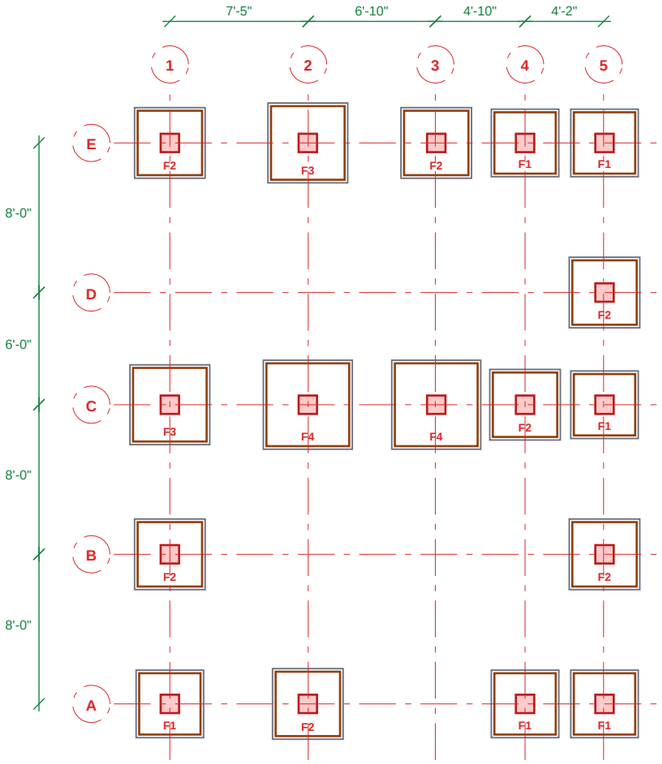
Project	**G+1 Residential House**
Project code	**PRJ-0027**
Client / owner	**Mr. Kumar**
Drawing	**Column Layout & Grid**
Sheet no.	**S-01**
Scale (on A3)	**1:75**
Concrete / steel	**M20 / Fe500**
SBC / seismic zone	**150 kN/m2 / Zone III**
Version / date	**v1 · 04-11-2026**
Prepared by (signature)	
Structural engineer (signature & reg. no.)	

PRELIMINARY DESIGN – NOT FOR CONSTRUCTION
until checked and signed by a qualified structural engineer.

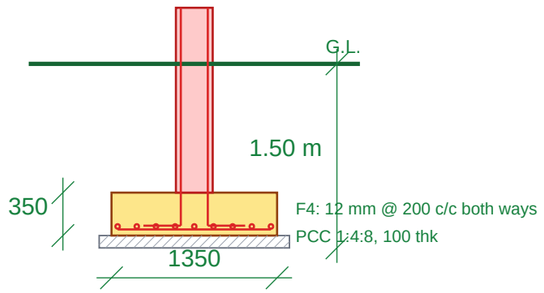
S-01

Foundation Plan & Footing Schedule

Mark	Size (m)	Depth (mm)	Steel	Column	Nos
F1	1.0 x 1.0	350	12 mm @ 200 c/c both ways	300x300	6
F2	1.05 x 1.05	350	12 mm @ 200 c/c both ways	300x300	7
F3	1.2 x 1.2	350	12 mm @ 200 c/c both ways	300x300	2
F4	1.35 x 1.35	350	12 mm @ 200 c/c both ways	300x300	2



FOUNDATION PLAN (isolated footings)



TYPICAL FOOTING F4 - SECTION (all sizes mm)

Praganya Construction

Proprietor: Nishanth S · 98xxxxxx

Project
G+1 Residential House

Project code
PRJ-0027

Client / owner
Mr. Kumar

Drawing
Foundation Plan & Footing Schedule

Sheet no.
S-02

Scale (on A3)
1:100

Concrete / steel
M20 / Fe500

SBC / seismic zone
150 kN/m2 / Zone III

Version / date
v1 · 04-11-2026

Prepared by (signature)

Structural engineer (signature & reg. no.)

PRELIMINARY DESIGN – NOT FOR CONSTRUCTION
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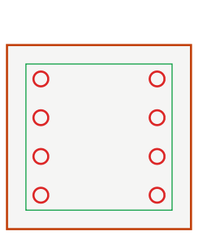
S-02

Column Schedule & Sections

Mark	Size (mm)	Main bars	Ties	Pu (kN)	Nos
C1	300 x 300	8-12 mm	8 mm @ 150 c/c	98.5	17

C1



300x300

8-12 mm

ties 8 mm @ 150 c/c

Column sections – scale 1:10 on A3

Praganya Construction

Proprietor: Nishanth S · 98xxxxxx

Project
G+1 Residential House

Project code
PRJ-0027

Client / owner
Mr. Kumar

Drawing
Column Schedule & Sections

Sheet no.
S-03

Scale (on A3)
see sections

Concrete / steel
M20 / Fe500

SBC / seismic zone
150 kN/m2 / Zone III

Version / date
v1 · 04-11-2026

Prepared by (signature)

Structural engineer (signature & reg. no.)

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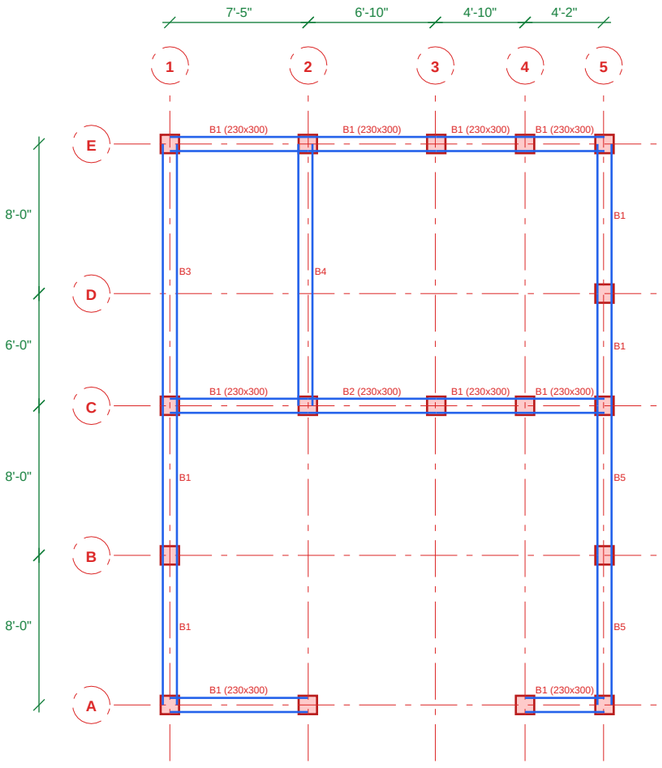
S-03

Beam Layout – Ground Floor roof

Mark	b x D	Bottom	Top	Stirrups
B1	230x300	2-12 mm	2-12 mm	2L-8 mm @ 175 c/c
B2	230x300	3-12 mm	2-12 mm	2L-8 mm @ 175 c/c
B3	230x400	3-16 mm	2-12 mm	2L-8 mm @ 250 c/c
B4	230x475	4-16 mm	2-12 mm	2L-8 mm @ 300 c/c
B5	230x300	4-12 mm	2-12 mm	2L-8 mm @ 175 c/c



BOTTOM 3-16 mm · stirrups 2L-8 @ 100 c/c for 2D at ends, 250 c/c middle

TYPICAL BEAM B3 (230 x 400)

Praganya Construction

Proprietor: Nishanth S · 98xxxxxx

Project
G+1 Residential House

Project code
PRJ-0027

Client / owner
Mr. Kumar

Drawing
Beam Layout – Ground Floor roof

Sheet no.
S-04a

Scale (on A3)
1:100

Concrete / steel
M20 / Fe500

SBC / seismic zone
150 kN/m2 / Zone III

Version / date
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Prepared by (signature)

Structural engineer (signature & reg. no.)

PRELIMINARY DESIGN – NOT FOR CONSTRUCTION
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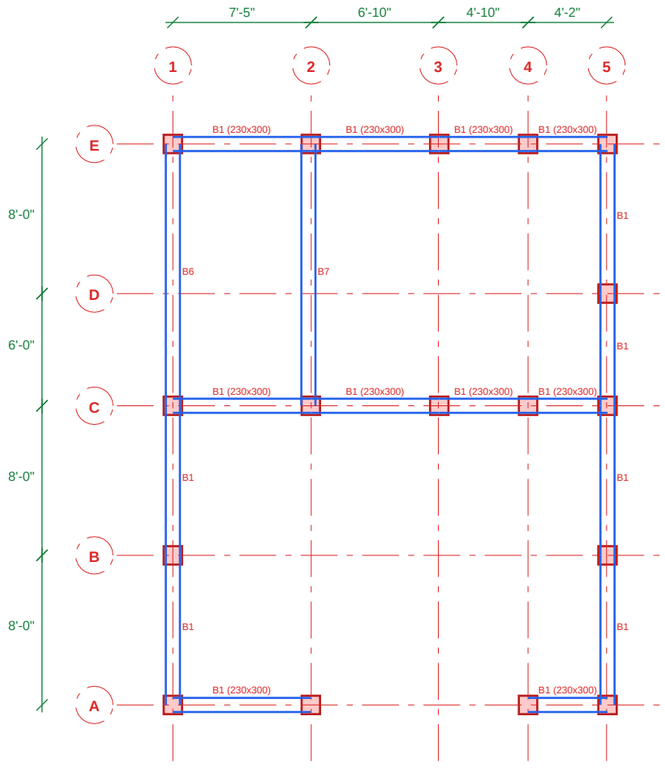
S-04a

Beam Layout – First Floor roof

Mark	b x D	Bottom	Top	Stirrups
B1	230x300	2-12 mm	2-12 mm	2L-8 mm @ 175 c/c
B6	230x375	4-12 mm	2-12 mm	2L-8 mm @ 250 c/c
B7	230x375	2-12 mm	2-12 mm	2L-8 mm @ 250 c/c



Praganya Construction

Proprietor: Nishanth S · 98xxxxxx

Project
G+1 Residential House

Project code
PRJ-0027

Client / owner
Mr. Kumar

Drawing
Beam Layout – First Floor roof

Sheet no.
S-04b

Scale (on A3)
1:100

Concrete / steel
M20 / Fe500

SBC / seismic zone
150 kN/m2 / Zone III

Version / date
v1 · 04-11-2026

Prepared by (signature)

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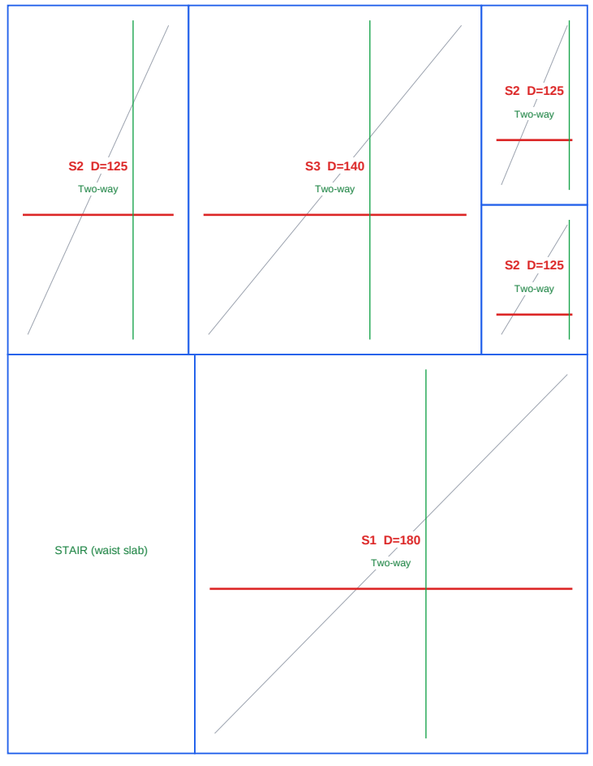
S-04b

Slab Layout – Ground Floor roof

Mark	Room	lx x ly (m)	D	Main (short)	Dist./long
S1	Car Parking	4.8 x 4.88	180	8 mm @ 230 c/c	8 mm @ 230 c/c
S2	Kitchen	2.21 x 4.27	125	8 mm @ 300 c/c	8 mm @ 300 c/c
S3	Living / Hall	3.58 x 4.27	140	8 mm @ 290 c/c	8 mm @ 290 c/c
S2	Passage / Wash	1.3 x 1.83	125	8 mm @ 300 c/c	8 mm @ 300 c/c
S2	Common Toilet	1.3 x 2.44	125	8 mm @ 300 c/c	8 mm @ 300 c/c



Praganya Construction

Proprietor: Nishanth S · 98xxxxxx

Project
G+1 Residential House

Project code
PRJ-0027

Client / owner
Mr. Kumar

Drawing
Slab Layout – Ground Floor roof

Sheet no.
S-05a

Scale (on A3)
1:75

Concrete / steel
M20 / Fe500

SBC / seismic zone
150 kN/m2 / Zone III

Version / date
v1 · 04-11-2026

Prepared by (signature)

Structural engineer (signature & reg. no.)

PRELIMINARY DESIGN – NOT FOR CONSTRUCTION
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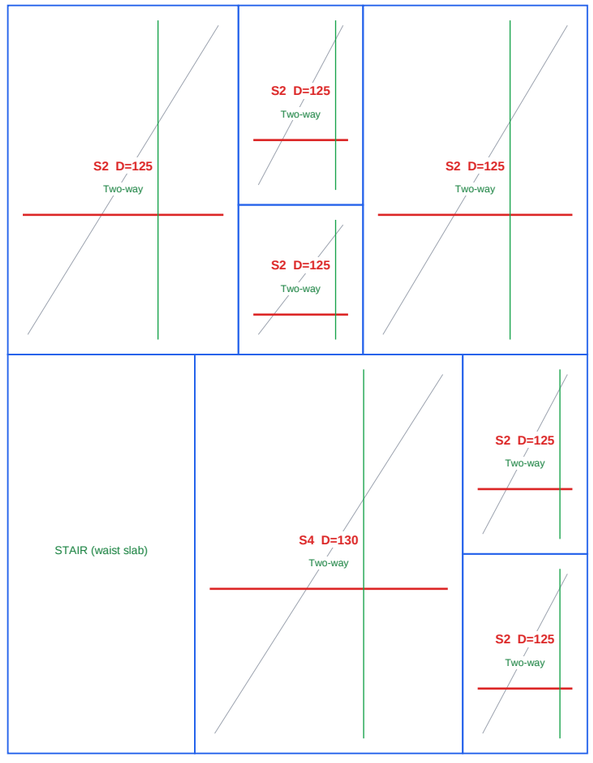
S-05a

Slab Layout – First Floor roof

Mark	Room	lx x ly (m)	D	Main (short)	Dist./long
S4	Family Lounge	3.28 x 4.88	130	8 mm @ 270 c/c	8 mm @ 300 c/c
S2	Passage / Wash	1.52 x 2.44	125	8 mm @ 300 c/c	8 mm @ 300 c/c
S2	Common Toilet	1.52 x 2.44	125	8 mm @ 300 c/c	8 mm @ 300 c/c
S2	Master Bedroom	2.82 x 4.27	125	8 mm @ 300 c/c	8 mm @ 300 c/c
S2	Attached Toile	1.52 x 2.44	125	8 mm @ 300 c/c	8 mm @ 300 c/c
S2	Dress	1.52 x 1.83	125	8 mm @ 300 c/c	8 mm @ 300 c/c
S2	Bedroom	2.74 x 4.27	125	8 mm @ 300 c/c	8 mm @ 300 c/c



Praganya Construction

Proprietor: Nishanth S · 98xxxxxx

Project
G+1 Residential House

Project code
PRJ-0027

Client / owner
Mr. Kumar

Drawing
Slab Layout – First Floor roof

Sheet no.
S-05b

Scale (on A3)
1:75

Concrete / steel
M20 / Fe500

SBC / seismic zone
150 kN/m2 / Zone III

Version / date
v1 · 04-11-2026

Prepared by (signature)

Structural engineer (signature & reg. no.)

PRELIMINARY DESIGN – NOT FOR CONSTRUCTION
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S-05b

General Notes, Checks & Quantities

- Concrete M20, steel Fe500. Covers: slab 20 mm, beam 25 mm, column 40 mm, footing 50 mm.
- Loads: live 2.0 kN/m² (roof 1.5), floor finish 1.0 kN/m², masonry 20.0 kN/m³, safe bearing capacity 150 kN/m² (given), foundation depth 1.5 m.
- Gravity loads only (IS 456). Wind / earthquake analysis is not part of this preliminary design.
- Lap length 50 d (tension), development length 47 d (Fe500, M20). Hooks 135° for ties and stirrups.
- PRELIMINARY DESIGN – to be checked and signed by a qualified structural engineer before construction.

Member	Concrete (m3)	Steel (kg, approx.)
Footings	7.09	425
Columns	11.63	2093
Beams	5.97	895
Slabs	15.01	1351
Total	39.69	4764

Praganya Construction

Proprietor: Nishanth S · 98xxxxxx

Project
G+1 Residential House

Project code
PRJ-0027

Client / owner
Mr. Kumar

Drawing
General Notes & Quantities

Sheet no.
S-06

Scale (on A3)
-

Concrete / steel
M20 / Fe500

SBC / seismic zone
150 kN/m2 / Zone III

Version / date
v1 · 04-11-2026

Prepared by (signature)

Structural engineer (signature & reg. no.)

PRELIMINARY DESIGN – NOT FOR CONSTRUCTION
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